

Assignment 2

Wind Speed Blade Angle Rotor Speed Power Output

5	11	125	48
7	9	155	72
9	15	175	101
11	13	220	142
13	18	235	176

$$y = C + m_1 x_1 + m_2 x_2 + m_3 x_3$$

$$\theta = (X^T X)^{-1} X^T y$$

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}$$

$$y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$A = X^T X = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8460 \\ 66 & 630 & 920 & 12485 \\ 910 & 8460 & 12485 & 143900 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$|A| = 5343750$$

$$\text{adj } A = \begin{bmatrix} 74062500 & 13218750 & 1811250 & -268125 \\ 13218750 & 3492187.5 & 272812.5 & 14531.25 \\ 1811250 & 272812.5 & 656718.75 & 26015.625 \\ -268125 & 14531.25 & 26015.625 & 2798.4375 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 13.8548 & -2.4746 & -0.1808 & 0.0502 \\ -2.4746 & 0.6531 & -0.0510 & -0.0027 \\ -0.1808 & -0.0510 & 0.1229 & -0.0049 \\ 0.0502 & -0.0027 & -0.0049 & 0.0005 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 4 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 7 \\ 125 & 155 & 145 & 220 & 235 \end{bmatrix} \begin{bmatrix} 48 \\ 42 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$\begin{bmatrix} 539 \\ 5563 \\ 7705 \\ 107435 \end{bmatrix}$$

$$\theta = (X^T X)^{-1} (X^T Y) = \begin{bmatrix} -99.9112 \\ 1.2174 \\ 2.4166 \\ 0.9058 \end{bmatrix} = \begin{bmatrix} c \\ m_1 \\ m_2 \\ m_3 \end{bmatrix}$$

$$y = -99.9112 + 1.2174 x_1 + 2.4166 x_2 + 0.9058 x_3$$

$$x_1 = 9$$

$$x_2 = 11$$

$$x_3 = 100$$

$$y = -99.9112 + 1.2174(9) + 2.4166(11) + 0.9058(100) = \underline{\underline{28.208}}$$